# ML-Driven Trading on Kalshi BTC Binary Markets

## Machine Learning Final Project

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The University of Chicago

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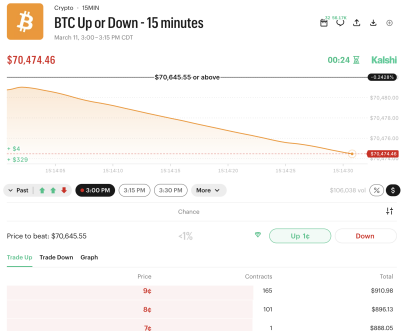
# Problem Statement

**The Exchange: Kalshi**

- CFTC-regulated prediction market exchange
- Binary contracts: pays \$1 if event occurs, \$0 otherwise

**BTC Up or Down — 15 Min**

- New contract every 15 min: "Will BTC be $\geq$ \$X at expiry?"
- YES/NO sides trade as probabilities (cents); live bid/ask order book
- Settles at \$1 (YES) or \$0 (NO)



# Data Sources

**Kalshi API (REST & WebSocket)**

- Top-of-book YES/NO: bid/ask price & size
- Trade stats: count, volume, VWAP (last 60s)
- Quote-change event count per minute
- Frequency: **20-second snapshots**, real-time

**Coinbase API**

- BTC/USD spot price, **every 20 seconds**, real-time

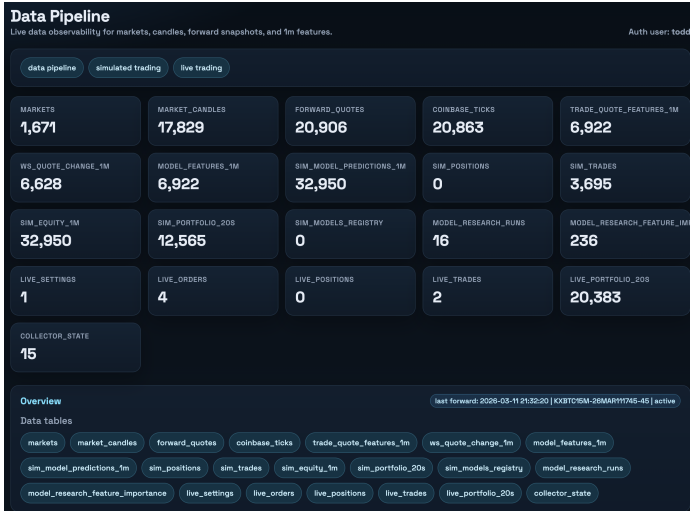
**Collection & Storage**

- Continuous collection: **Mar 6, 2026, 7:16 PM CST – present**
- MySQL on AWS RDS; 20s snapshots aggregated to 1-min bars
- EC2 workers: `forward_collector`, `coinbase_collector`, `feature_1m_collector`

**Key Tables**

- `forward_quotes` — 20s top-of-book
- `coinbase_ticks` — 20s BTC spot
- `model_features_1m` — 1-min ML feature table

# Data Pipeline Dashboard



# Feature Set

### Order Book & Pricing

- `yes_bid/ask_dollars` — top-of-book prices
- `spread_dollars`, `mid_dollars`
- `orderbook_imbalance` — $(bid_sz - ask_sz)/(bid_sz + ask_sz)$

### Contract Returns

- `mid_ret_1m`, `mid_ret_3m`

### Trade Activity (last 60s)

- `trades_last_60s_count` — trade count
- `trades_last_60s_volume_fp` — dollar volume
- `trades_last_60s_vwap_yes` — VWAP
- `quote_change_events_60s` — book update frequency

### Coinbase BTC Spot

- `cb_ret_1m`, `cb_ret_3m`, `cb_ret_5m`
- `cb_vol_5m` — 5-min realized volatility

### Moneyness & Interactions

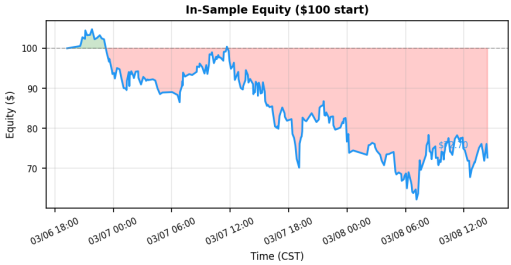
- `moneyness_abs/pct` — spot vs. strike
- `moneyness_vol_scaled` — $\div$ BTC vol
- `moneyness_x_sqrt_tte` — $\times\sqrt{TTE}$
- `moneyness_x_imbalance` — $\times$ imbalance
- `moneyness_x_qchg` — $\times$ quote changes

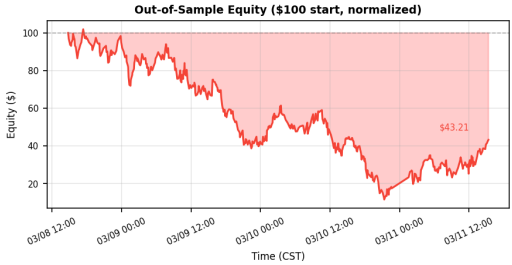
# Model Baseline — Naive Logistic Regression

**Label:** $y_t = \mathbf{1}[\text{bid}_{t+1} > \text{ask}_t]$ **Features (13):** `bid, ask, spread, mid, imbalance, mid_ret_1m/3m, cb_ret_1m/3m, trades_count, quote_chg, tte, moneyness_pct`

| | | | |
|---|---|---|---|
| **Train (CST)** | Mar 6 7:16 PM – Mar 8 6:16 AM | 2,044 rows | In-sample |
| **Test (CST)** | Mar 8 6:16 AM – 2:47 PM | 511 rows | In-sample holdout |
| **OOS (CST)** | Mar 8 2:47 PM – present | live | Forward test |

**Buy YES:** $p_{\text{yes}} \geq 0.55$ & edge $> \$0.01$ **Sell YES:** $p_{\text{yes}} < 0.55$ or edge $\leq \$0.01$ (or TTE $\leq 60$s)
**Qty:** 1–10 contracts (scaled by edge) **OOS PnL: –$56.79** / 474 trades / 49.7% win rate



# Model #1 — Logistic Regression with L1 and Interactions

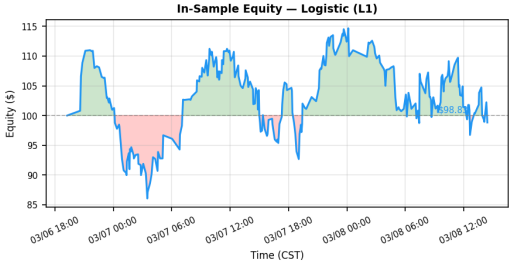
**Label:** $y_t = \mathbf{1}[\text{bid}_{t+1} > \text{ask}_t]$ **Features (22):** all baseline features + `cb_ret_5m`, `cb_vol_5m`, `moneyness_abs`, `moneyness_vol_scaled`, `moneyness_x_sqrt_tte`, `moneyness_x_imbalance`, `moneyness_x_trades_volume_fp`, `trades_vwap`

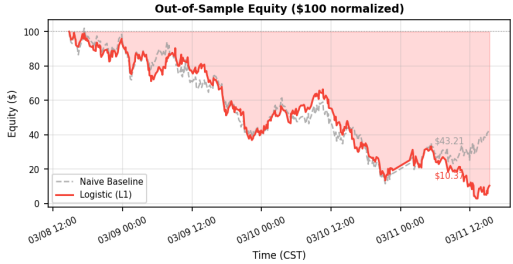
**Regularization:** L1 ($\lambda = 2.0$)

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**OOS PnL: –$89.63** ($100→$10.37) / 501 trades / 46.3% win rate

**Comment:** Added complexity is not justified; near-random signal (∼50/50) motivates non-linear models



# Model #2 — Gradient Boosting

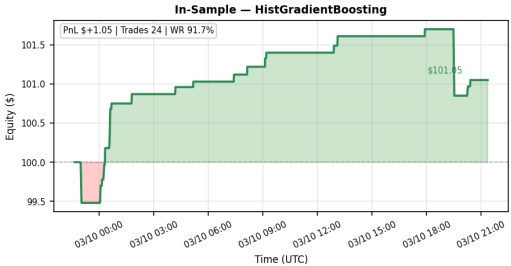
**Label:** $y_t = \mathbf{1}$[final YES bid $\geq$ \$0.90] (expiry-resolution) **Features (39):** all 22 production features + depth/pressure, flow context, return interactions, strike carry

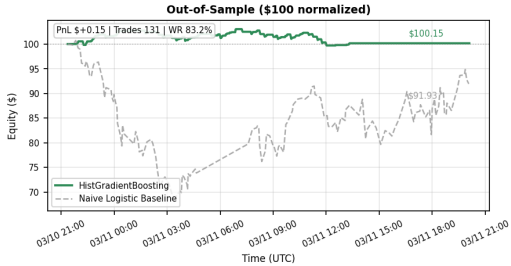
**Params:** `learning_rate=0.03, max_depth=4, max_iter=250, min_samples_leaf=25 l2=0.01, max_leaf_nodes=63`

| | | | |
|---|---|---|---|
| **Train** | Mar 7 – Mar 9 (UTC) | 3,545 rows | 60% split |
| **Validation** | Mar 9 22:38 – Mar 10 21:21 (UTC) | 1,182 rows | In-sample holdout |
| **OOS (Test)** | Mar 10 21:22 – Mar 11 20:06 (UTC) | 1,182 rows | Forward test |

**Buy:** $p_{\text{yes}} \geq 0.55$ & edge $\geq$ \$0.08 **Exit:** take-profit at entry +\$0.05, else final YES bid **Size:** 1

**OOS PnL: +\$0.15** / 131 trades / 83.2% win rate **OOS AP:** 0.814, **AUC:** 0.810



# Model #3 — Random Forest

**Label:** $y_t = \mathbf{1}$[final YES bid $\geq$ \$0.90] (expiry-resolution) **Features (39):** all 22 production features + depth/pressure, flow context, return interactions, strike carry

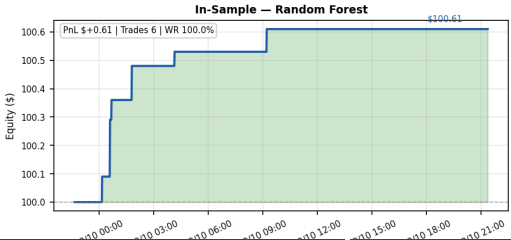
**Params:** `n_estimators=500, max_depth=10, min_samples_leaf=4, min_samples_split=12, max_features=sqrt, class_weight=balanced_subsample`

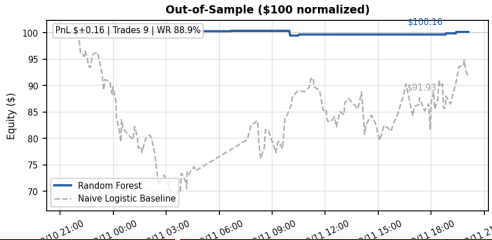
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| **OOS (Test)** | Mar 10 21:22 – Mar 11 20:06 (UTC) | 1,182 rows | Forward test |

**Buy:** $p_{\text{yes}} \geq 0.65$ & edge $\geq$ \$0.08 **Exit:** take-profit at entry +\$0.05, else final YES bid **Size:** 1

**OOS PnL: +\$0.16** / 9 trades / 88.9% win rate **OOS AP:** 0.848, **AUC:** 0.847

**Comment:** selectivity and expiry-resolution label are the key drivers



# Conclusion

# Thank You

Questions & Discussion